DATA ANALYSIS QUESTIONS

Graduation Project - DEPI Data Analysis Track

MANUFACTURING DOWNTIME ANALYSIS

Overview

This document lists the business questions and the five scopes of questions that the Manufacturing Downtime analysis must answer. The answers come from the Power BI dashboard, and the last scope uses a Python forecast.

- **Data:** 38 batches over 4 production days (29, 30, 31 Aug and 2 Sep 2024), 6 products, 4 operators and 11 downtime factors.
- **Tables:** *Line productivity* (one row per batch) and *Line downtime* (one row per batch and factor).
- **Tools:** Power Query for cleaning, Power BI for the dashboard, Python for the forecast.

Part	Name	Questions	Focus
Business Questions	Management-level questions	7	High-level questions the analysis answers overall
Scope 1	Production Performance Analysis	8	Batches, durations and the gap to the theoretical minimum time
Scope 2	Downtime Analysis	10	How much time is lost, by batch, factor and date
Scope 3	Root Cause Investigation	10	Which factors matter most, operator errors, and links to products and operators
Scope 4	Production Efficiency Analysis	10	Efficiency and downtime by product and operator, outliers and correlation
Scope 5	Predictive Analysis and Production Planning	2	Next-day downtime prediction and the number of batches that fit

Business Questions

#	Question
1	Which downtime factors should management prioritize to reduce production interruptions?
2	Which products experience the greatest downtime and longest production durations?
3	How did downtime vary across the recorded production dates?
4	How much recorded downtime was associated with factors classified as operator errors?
5	Which downtime factors may be contributing to longer production durations?
6	Are batches with higher downtime also associated with longer production durations?
7	Which operator-related downtime patterns warrant further investigation?

Scope 1: Production Performance Analysis — 8 Questions

#	Question
1	How many production batches were recorded during the available period?
2	What was the average actual production duration per batch?
3	Which batches had the longest actual production duration?
4	Which products had the highest average production duration?
5	Which batches had an actual production duration greater than their product's theoretical minimum batch time?
6	By how many minutes did each batch's actual production duration exceed or fall below its product's theoretical minimum batch time?
7	How did the number of recorded production batches vary across production dates?
8	How did average production duration differ by product flavor and size?

Scope 2: Downtime Analysis — 10 Questions

#	Question
9	What was the total recorded downtime across all production batches?
10	What was the average total downtime per batch?
11	Which batches had the highest total downtime?
12	Which downtime factors contributed the most total recorded downtime?
13	Which downtime factors had the highest average downtime per affected batch?
14	Which downtime factors affected the greatest number of batches?
15	What percentage of total recorded downtime was contributed by each downtime factor?
16	Which batches contributed the largest share of total recorded downtime?
17	How did total recorded downtime vary across production dates?
18	Which products had the highest average downtime per batch?

Scope 3: Root Cause Investigation — 10 Questions

#	Question
19	Which downtime factors collectively accounted for approximately 80% of total recorded downtime?
20	Which downtime factors occurred most frequently across production batches?
21	How many minutes of downtime were associated with factors classified as operator errors?
22	What percentage of total recorded downtime was associated with operator-error-related factors?
23	Which operator-error-related factors contributed the most downtime?
24	Which downtime factors contributed the most downtime for each product?
25	Were particular downtime factors associated with batches handled by particular operators?
26	Were particular downtime factors associated with longer actual production durations?
27	Which downtime factors remained prominent across multiple production dates?
28	How did average production duration and downtime differ across operators?

Scope 4: Production Efficiency Analysis — 10 Questions

#	Question
29	Which products had the largest average production-duration overruns compared with their theoretical minimum batch times?
30	Which products had the highest average production duration relative to their theoretical minimum batch times?
31	Which products had the highest average downtime per batch?
32	Which operators' batches had the highest average downtime per batch?
33	How did average production duration differ across operators?
34	How did downtime vary between products of different flavors and sizes?
35	Which products had the highest percentage of batches exceeding their theoretical minimum batch time?
36	Which batches had unusually high downtime compared with the overall batch average?
37	Were higher downtime levels associated with larger differences between actual production duration and theoretical minimum batch time?
38	How did the proportion of downtime attributed to different factors vary across production dates?

Scope 5: Predictive Analysis and Production Planning — 2 Questions

#	Question
39	Can historical production and downtime data be used to predict downtime for the next day of operation?
40	Based on the predicted downtime and available production time, how many batches could theoretically be produced during the next day of operation?